Source spoofing and reflection

Follow-along: play the attacker, then the defender

Overview

Definition: Source-address spoofing

Forging the source-address field of an outgoing packet so it names an address the sender does not own. UDP carries no handshake, so nothing stops a host from claiming any source; the receiver, and any reply it sends, treats the forged address as genuine.

Definition: Reflection

Bouncing traffic off a third-party server instead of sending it to the target directly. The attacker sends a request with the victim's address forged as the source; the server answers the source it was handed, so its reply lands on the victim, which never asked.

Source-address spoofing and **reflection** turn ordinary public servers into a way to reach someone who never contacted you. The attacker never sends to the victim directly: it sends a request to a server that answers over UDP, and forges the request's source so the answer is delivered to the victim. Because a UDP server trusts the source field and replies to whoever the packet claims to be from, the server does the delivering, and its own address, not the attacker's, is what the victim sees.

Spoofing is the root cause. The same forged source that steers a server's reply onto the victim also keeps the attacker's own address out of everything the victim sees. This lab builds reflection from two everyday services, DNS and NTP, then closes the single hole it depends on: a router that forwards packets without checking whether a source address could legitimately have arrived where it did.

Running this lab in the TUI

You drive the lab from the MiniLabs launcher. Clone it if the machine does not have it, and pull if it does, because handouts and lab scripts change together:

```
git clone https://github.com/nzRichie/MiniLab.git ~/MiniLab   # first time
cd ~/MiniLab && git pull                                      # already cloned
```

Start the launcher from that directory:

```
cd ~/MiniLab && ./minilabs
```

If it exits with a Docker error, follow the setup in the repository's `README.md` and run it again.

The root menu lists one lab per row. Select **Source spoofing and reflection** to open its actions:

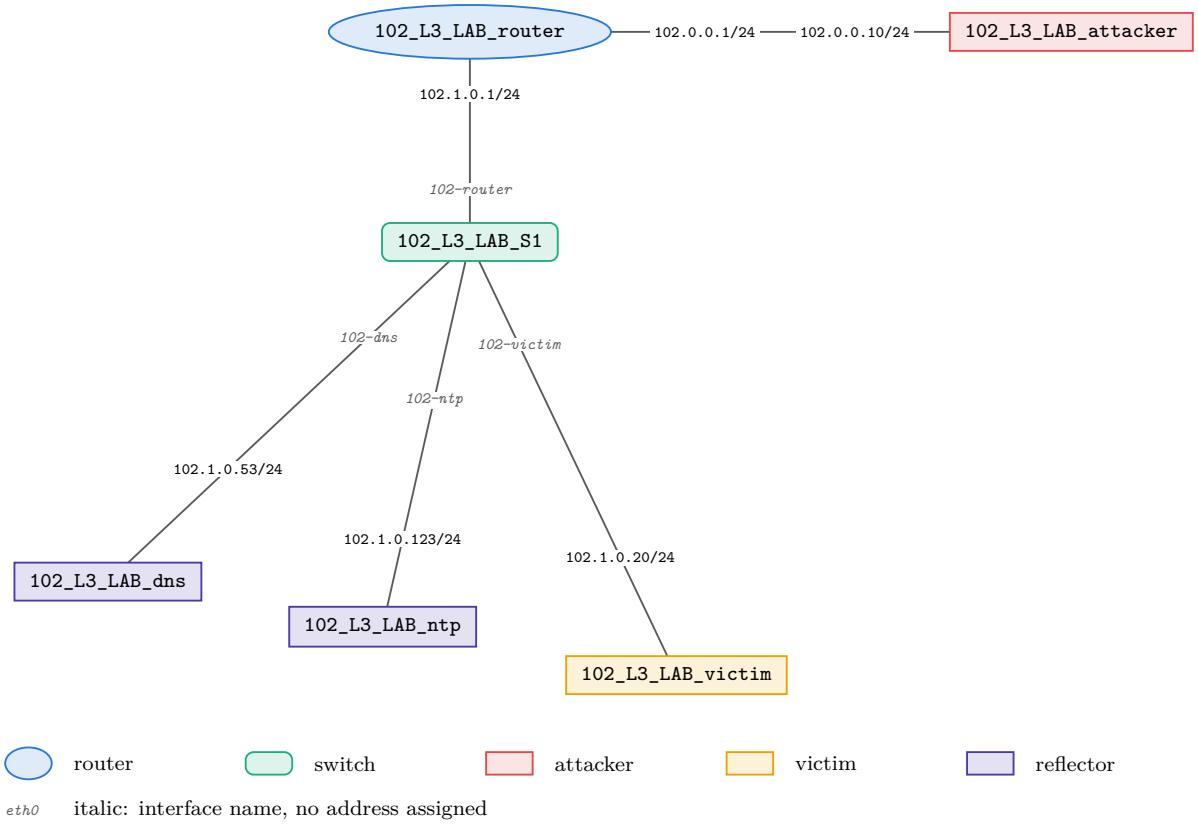
- **Spawn** creates this lab's router, switch, attacker, victim, and two reflectors, and wires them. Wait for the panel to turn green before you go on. The first spawn of a lab also builds its images, which takes a few minutes; later spawns take seconds.

- **Status** prints the addressing, the router's filter state, and the oracle: does a forged-source query still reach each reflector (REACHES) or not (BLOCKED). Status takes over the whole screen: the arrow keys scroll its output, and Enter on **Done** closes it.
- **Shell command for a node** asks which node (attacker, router, victim, dns, ntp) and prints the `docker exec -it ...` line to paste into your own terminal; each step below names the node it runs on.
- **Reset to baseline** stops the attack and drops your filter, returning the lab to its clean Part 1 state without tearing it down.
- **Teardown** removes the lab.
- **Open handout** opens this document.

Spawn the lab now, then work through Part 1 in your own terminal. **Back** (or Esc) returns to the lab list.

The setup

This lab is Layer 3, so it is not one flat LAN. A single router forwards between two subnets, and everything the attack and the defence do happens across that router:



Each node is labelled with the container name you pass to `docker exec`, and each end of a link with the address on that interface. The same addressing, to copy from:

Role	IP	Subnet	Container
attacker	`102.0.0.10`	customer `102.0.0.0/24`	`102_L3_LAB_attacker`
router	`102.0.0.1` (cust) `102.1.0.1` (core)	both	`102_L3_LAB_router`
victim	`102.1.0.20`	core `102.1.0.0/24`	`102_L3_LAB_victim`
DNS reflector	`102.1.0.53`	core	`102_L3_LAB_dns`
NTP reflector	`102.1.0.123`	core	`102_L3_LAB_ntp`

The attacker sits on the customer edge, on a point-to-point link to the router's `cust` interface. The victim and the two reflectors share the core segment behind the router's `core` interface, on one Open vSwitch switch. The attacker's NIC is `102-cust`; every core host's NIC is `102-S1`.

The attacker carries stock tooling; there is no bespoke exploit:

- `nping` (from the nmap project): sends UDP or ICMP with a source address you choose (`-S`). This is how you forge a source.
- `scapy` (Python): builds the reflected DNS and NTP queries field by field, including the forged source. You type every field, so you see exactly what leaves.
- `reflect`: a baked one-command version of the same two queries. It takes no default addresses; you still name the reflector, the protocol, and the source you forge.
- `tcpdump`, `dig`: read what arrives at the victim and confirm the reply lands there rather than on the attacker.

The router forwards between the two subnets and does no source checking yet. That gap is the attack surface; closing it is Part 2. Open a shell wherever a step says so with:

```
docker exec -it <container> bash
```

Assessment. This lab is graded on thirteen questions, marked inline below as **Question** *N*, each placed at the step whose output it asks about. Answer all thirteen in a separate document as you work through the steps. Every question is answered from what you see on screen at that step.

Part 1: the attack

Honest traffic first

Definition: TXT record

A DNS record type that holds arbitrary text rather than an address. Domains publish TXT records for policy and verification data; one name can carry many TXT strings, and a query for that name returns the whole set. A set too large for one UDP datagram comes back truncated instead, with the TC bit set so the client retries over TCP; `reflect.lab` holds one short string, so this one fits.

1. **Send one query the honest way.** On the attacker, ask the DNS reflector for its one record, using your own address as the source:

```
docker exec -it 102_L3_LAB_attacker bash
dig @102.1.0.53 reflect.lab TXT        # a short TXT record comes back
```

The reply reaches you because the query carried your real source, `102.0.0.10`, so the reflector returns the **TXT record** to you. The only thing that decides where the reply goes is the source field you sent, which is the hinge the rest of the lab turns on.

Question 1. You sent this query from your own address and the reply came back to you. Predict what changes about where the reply goes if the source field named the victim (`102.1.0.20`) instead, and explain what makes UDP reflection possible that a TCP-based service would not allow.

Forge a source

2. **Send a packet from an address that is not yours.** Still on the attacker, aim UDP at the victim but claim a source on no lab subnet (`203.0.113.66`, TEST-NET-3). In a second shell, watch the victim receive it:

```
docker exec -it 102_L3_LAB_victim bash
tcpdump -n -i 102-S1 'src host 203.0.113.66'     # leave running
```

```
# back on the attacker:
nping --udp -S 203.0.113.66 -p 9 -c 4 --rate 5 -e 102-cust 102.1.0.20
```

The victim's `tcpdump` shows four packets from `203.0.113.66`. The router forwarded them from `cust` to `core` without asking whether that source could legitimately be there.

Question 2. The victim received four packets sourced from `203.0.113.66`, an address no host owns. State what the router did with each packet and which check it did not perform.

Question 3. Any reply the victim sends goes to `203.0.113.66`, not to you. Name one thing forging the source already buys the attacker, before any reflection is involved.

Reflect off DNS

3. **Point a reflector's reply at the victim.** Now forge the source as the *victim* and query the DNS reflector. The reflector will answer the victim, which never asked. Start the capture *first* and leave it running; the network is otherwise idle, so what it catches is the attack:

```
# on the victim:
tcpdump -ne -i 102-S1 'src host 102.1.0.53 and dst host 102.1.0.20'
```

Then, on the attacker, build the query by hand in scapy so every field is yours, forging the victim as the source:

```
# on the attacker:
python3 -c "from scapy.all import *; \
  send(IP(src='102.1.0.20', dst='102.1.0.53')/UDP(dport=53)/ \
       DNS(rd=1, qd=DNSQR(qname='reflect.lab', qtype='TXT')), count=4,
    inter=0.2)"
```

The baked tool sends the same packet: `reflect dns 102.1.0.53 102.1.0.20 4`. On the victim's capture, four replies arrive from the DNS server, each carrying the TXT record the reflector serves, addressed to a host that never asked for it.

Question 4. The query's source was set to `102.1.0.20`. Explain why the victim, not the attacker, is the one that receives the replies.

Reflect off NTP

Definition: NTP control query (mode 6)

A management request to an NTP server, separate from ordinary time synchronisation. A mode-6 `readvar` asks the server to return its whole internal variable list in one reply.

4. **Reflect off a second, unrelated service.** Reflection is not specific to DNS: any UDP service that answers whoever a request claims to be from can bounce traffic at a victim. NTP is a different protocol on a different port, and the same forged-source trick works unchanged. NTP's original reflection vector was the mode-7 `monlist` command, removed from `ntpd` in 4.2.7p26 in 2010; `ntpsec`, which this lab runs, answers no mode-7 request at all. The vector that replaced it is a mode-6 `readvar` **control query**, which returns the server's whole variable list. Start the capture on the victim *first*:

```
# on the victim:
tcpdump -ne -i 102-S1 'src host 102.1.0.123 and dst host 102.1.0.20'
```

Then fire the query from the attacker, forging the source as the victim again:

```
# on the attacker:
python3 -c "from scapy.all import *; \
  send(IP(src='102.1.0.20', dst='102.1.0.123')/UDP(dport=123)/ \
       Raw(b'\x16\x02'+b'\x00'*10), count=4, inter=0.2)"
```

The baked equivalent is `reflect ntp 102.1.0.123 102.1.0.20 4`. The twelve bytes `16 02 00 ...` are a mode-6 control header built by hand: `16` carries version 2 and mode 6 (control), `02` is opcode 2 (read variables), and the ten zero bytes are the sequence, status, association, offset and count fields. A byte too many or too few and the server treats it as malformed and returns almost nothing. (A real `ntpq -c rv` is close but not identical: `ntpsec`'s client sets the leap-indicator bits, sending `d6` for that first byte, and starts the sequence field at 1.)

Question 5. You reflected off two unrelated services, DNS and NTP, with the same forged-source packet. State the one property a service must have for reflection to work off it, then decide whether a UDP echo service on port 7 (which replies with a copy of whatever it receives) could be used the same way, and justify.

Why bounce it at all

5. **See the attacker's position.** At the victim, the flood arrives from `102.1.0.53` and `102.1.0.123`: two legitimate servers. The attacker's own address, `102.0.0.10`, appears nowhere in what the victim sees.

Question 6. Give two reasons an attacker bounces traffic off reflectors instead of sending it straight to the victim. Tie each reason to a source address in the victim's capture, not to the volume of traffic.

Question 7. The victim sees the flood coming from two real servers it may depend on. Explain why blocking those source addresses at the victim is not a usable fix.

Part 2: the defence

The attack works only because the router forwards a packet without asking whether its source address belongs where it arrived. The defence closes exactly that gap, and it goes on the router at the *attacker's* edge, not near the victim: the network a packet leaves is the only one that can check any source address it sees, because only it knows which addresses live behind that interface. A network further away can still reject the narrow case of a packet arriving from outside that carries one of its own addresses as the source, but only the first hop can judge every source. Both mechanisms below enforce the same one rule at the customer interface.

Definition: Source-address verification

Checking, at the interface a packet arrives on, that its source address is one that could legitimately originate there, and dropping it otherwise. It is applied at the edge closest to the sender, not near the target.

Definition: BCP 38 (ingress filtering)

The IETF best-practice document that tells a network to drop, at its customer edge, any packet whose source address does not belong to that customer. It is the standard behind source-address verification.

The victim is drowning, but the fix does not go near the victim. It goes on the router at the *attacker's* edge, on the `cust` interface, and it is one idea: a packet arriving on an interface must carry a source address that could legitimately come from that interface. This is **source-address verification**, the **ingress filtering** of **BCP 38**. Try either mechanism below; both run on the router.

Question 8. The flood lands on the victim, yet the filter belongs on the router beside the attacker. Explain why a filter at the victim's edge cannot stop a reflection attack, and what the source-side filter stops that the victim-side one cannot.

Option A: strict uRPF on the router

Definition: uRPF (unicast reverse-path forwarding)

A per-packet check that consults the routing table: for a packet's source address, would the route back to it leave by the interface the packet just arrived on? In strict mode, if the answer is no, the packet is dropped.

Unicast Reverse Path Forwarding (uRPF) asks the routing table a question for every incoming packet: if I had to reach this packet's *source*, would I send it back out the interface it just arrived on? If not, drop it. Turn it on for the customer interface (the router runs privileged, so `/proc/sys` is writable here):

```
docker exec -it 102_L3_LAB_router bash
sysctl -w net.ipv4.conf.all.rp_filter=0     # the higher of all and cust is what
    applies
sysctl -w net.ipv4.conf.cust.rp_filter=1    # 1 = strict
```

The kernel takes the maximum of `all` and the per-interface value, so setting `all` to 0 leaves `cust` to decide. It is not a precedence rule: with `all` at 2 the interface would run loose whatever `cust` said.

Re-run every attack from Part 1. The forged-source packets stop arriving: the victim's `tcpdump` goes quiet, and the reflectors receive nothing to reflect. Confirm honest traffic is untouched by running the honest `dig` from Part 1 again; it still answers.

Question 9. Apply strict uRPF to the DNS reflection step's forged packet: source `102.1.0.20` arriving on `cust`. Say which route lookup uRPF performs on it and why that lookup makes the packet fail.

Question 10. `rp_filter=1` is strict; `rp_filter=2` is loose (the source need only be reachable by *some* route, not necessarily out the ingress interface). Describe one legitimate topology where strict mode would wrongly drop real traffic and loose mode would not.

Option B: an nftables ingress ACL

Definition: ACL (access control list)

An ordered list of match-and-action rules a device applies to packets. Here it permits sources inside the customer subnet and drops the rest, naming the allowed range literally rather than deriving it from the routing table as uRPF does.

An **ACL** states the same policy without consulting the routing table: on `cust`, permit only sources inside the customer subnet, drop the rest.

First undo Option A (`sysctl -w net.ipv4.conf.cust.rp_filter=0`), then, on the router:

```
nft -f - <<'EOF'
table ip bcp38 {
    chain ingress {
        type filter hook prerouting priority -300; policy accept;
        iif "cust" ip saddr != 102.0.0.0/24 counter drop
    }
}
EOF
```

Re-run the attacks; they are dropped again. This time you can read how many packets the rule caught:

```
nft list table ip bcp38        # the drop rule's counter climbs with each forged
    packet
```

Reference solution: `solution/nftables_acl.sh`.

Question 11. The rule drops when `iif "cust"` and `ip saddr != 102.0.0.0/24`. Explain why the attacker's honest traffic (source `102.0.0.10`) passes while every forged source you sent

is dropped.

Question 12. uRPF reads the routing table; the ACL names the subnet literally. Give one operational advantage of each: what uRPF handles for free that the ACL must be told, and what the ACL makes explicit that uRPF leaves implicit.

What this fixes, and its limit

Question 13. Suppose every operator ran this filter at its own customer edge. Of the two attack stages here, direct spoofing and reflection, state which stop working and why. Then explain why one network skipping the filter still leaves victims elsewhere exposed.

Checks: what "done" means

- **Attack works:** with no filter, the victim's `tcpdump` shows the forged-source packets and reflected replies from `102.1.0.53` and `102.1.0.123` it never asked for.
- **Defence holds:** with uRPF or the ACL on `cust`, the forged-source packets never leave the router, the victim's capture is empty, and the attacker's honest `dig` still answers.

`scripts/status.sh` fires one forged query at each reflector and reports REACHES or BLOCKED, so you can read before and after without shelling in. To undo your work, `scripts/reset.sh` stops the attack and drops the filter, back to the Part-1 baseline. `scripts/teardown.sh` removes the lab.

Safety. Everything here lives on one isolated router and switch with no route off the host. The forged packets and the reflected replies only ever reach the lab's own containers.

When you are done, tear the lab down. Choose **Teardown** in the TUI, or run `scripts/teardown.sh` directly, to remove the containers and their wiring so nothing is left running on your host. **Reset to baseline** only rewinds the attack and the defence; only **Teardown** deletes the lab.